



TIFAC-CORE



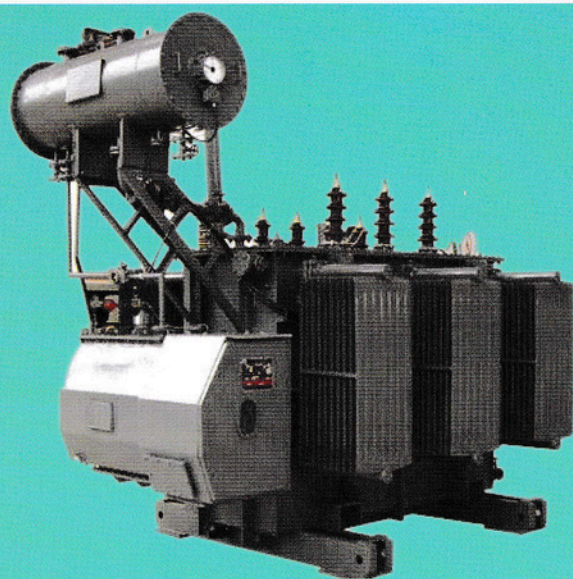
CENTRE FOR RELEVANCE AND EXCELLENCE

ON

POWER TRANSFORMER DIAGNOSTICS

VISION

TO CREATE A DULY ACCREDITED STATE OF THE ART "POWER TRANSFORMER CONDITION MONITORING AND DIAGNOSTIC CENTRE" TO CATER TO THE NEEDS OF INDUSTRY AND ACADEMIA.



NATIONAL INSTITUTE OF TECHNOLOGY HAMIRPUR (H.P.)

<http://tifac.nith.ac.in>

MISSION REACH

Mission REACH is an offshoot of 'Technology Vision 2020' program initiated by Late Dr. A.P.J. Abdul Kalam - Former President of India through Technology Information Forecasting and Assessment Council (TIFAC), Department of Science & Technology, Govt. of India Delhi to make India a developed nation by the year 2020. The program envisages to create a constellation of world class Centres Of Relevance & Excellence (COREs) in diverse disciplines across the country by the upgradation of science and technical institutions, mandated to produce technical manpower of international standard in the area of targeted excellence and tailor made to the emerging needs of industry. It is the befitting combination of the three most powerful pillars of economic growth i.e. industry, academic institutions and the government that provides the said mission, much needed strength and success. As on today, about 37 Centres of Relevance and Excellence (CORE) have been created in various technical institutes throughout the country in diverse areas of science and technology. NIT Hamirpur has the privilege of having one such TIFAC-CORE in area of Power Transformer Diagnostics that is functional wef 2004.

OBJECTIVES

- To develop CORE expertise in the chosen area to cater to the emerging needs of industry and academia in the form of skilled manpower in the targeted area to assist electrical utilities & carry out academic activities.
- To offer more elective courses for study to benefit UG students of electrical engineering department.
- To strengthen existing PG Program on Condition Monitoring Control and Protection of Electrical Apparatus of NIT Hamirpur.
- To meet the training/retraining needs of manpower engaged in power utilities all over the country in the area of power transformer diagnostics.
- To create experimental facilities & infrastructure of world class for quality research by sponsored R & D projects
- To establish linkage with pioneer organizations working in focused area all over the world

TIFAC-CORE IN POWER TRANSFORMER DIAGNOSTICS

The inception of TIFAC centre at NIT (Formerly Regional Engineering College, Hamirpur) is an outcome of relentless efforts by former Institute authorities, faculty of EED especially Prof. M. N. Bandyopadhyay (Advisor), Prof. Y.R. Sood (Chief Coordinator) & Dr. R.K. Jarial (PI & Coordinator) who have worked sincerely under Visionary leadership and mentorship provided by Prof. Chandra Shaker, the former Chairman, BOG NITH & Former Director NIT Hamirpur, Dr. R. L. Chauhan, former Chairman BOG, NIT Society and Principal, REC Hamirpur and other authorities of institute from time to time. For achieving coveted goal of self sustenance, the timely key support provided by Prof. Lalit K Awasthi, Former Director, NIT Hamirpur and Prof. H. M. Suryawanshi - the present Director, NITH would go a long way in rejuvenating further growth of TIFAC Centre in near future. Since Nov. 2000, the proposal envisaged a budget of Rs. 530 Lakhs to create TIFAC-CORE in "Power Transformer Diagnostics" that became functional wef 1st Sept 2004 after formal signing of MOU between TIFAC through Dept. of Science & Technology, Govt. of India Delhi & NIT Hamirpur on 10th Feb 2004 by Prof. Chandra shaker- the then Director, NITH. Since then, this Centre has been making every stride to enable its presence felt among leading Electric Power Utilities, IITs and similar reputed organizations in the country & abroad. This Centre has been working under mentorship of Apex body RMC having domain experts drawn from leading industries and academia. It has been seeking valuable suggestions through regular interactions with industry to make a mark in the area of on-line and off-line monitoring of electrical power apparatus especially power transformer, by creating infrastructure facilities of International Standards at TIFAC-CORE, NIT Hamirpur. To meet major objectives set under Mission REACH program of Govt. of India, consistent & continuous efforts are being made through kind support from its industrial partners including Himachal Pradesh State Electricity Board Limited, Shimla. An MOU to this effect was also signed between HPSEBL, Shimla and NIT Hamirpur in January 2006 by Prof. I K Bhat- former Director, NITH. To achieve set objectives, all possible efforts have been made by all faculty, EED and past Heads, EED, Prof. Y R Sood, Prof. Sushil Chouhan, Prof. R.N. Sharma, Dr. R. Nath Sharma and as Co-ordinator, TIFAC Centre respectively. Every one has made nice effort to develop the TIFAC entity at NITH to present level centre of relevance and excellence.

RELEVANCE OF TIFAC-CORE FOR ELECTRICAL UTILITIES

Power Transformer is known to be one of the most important and costliest equipment that is required in an Electric Power system. It plays a pivotal role in facilitating smooth supply of Electric Power through a challenging transmission network from a Generating Power House to Electrical Distribution Points. Their continuous loading operation & in service conditions may sometime cause outages and catastrophic failures. Transformer failure statistics reported in literature indicate that most failures have occurred before reaching their expected designed life of 40 years. Transformer failures are mainly related to transformer health conditions. Problems related to dielectric are reported as high as 75 percent. As per 2001 survey report submitted by a CIGRE (PARIS) working group on Failures Trends in large power transformers, about 41 % of failures were due to on-load tap changers (O.L.T.C) & about 19 % were attributed to the windings. The failure origins were 53% mechanical and 31% dielectric. Some power transformers without having on-load tap changers have 26.6 % of failures that were due to the windings, 6.4 % were due to the magnetic circuit, 33.3 % were due to terminals, 17.4 % were due to the tank and dielectric fluid, 11 % were due to other accessories and 4.6 % were due to misc reasons. The cost, time involved to repair and replacement of a power transformer is very high. The prolonged shutdown periods after catastrophic failures can be minimized in case electrical utilities opt for condition based maintenance program and not periodic maintenance program in respect of all major equipment's especially power transformer.

PG & PH.D. PROGRAMS AT TIFAC-CORE

PG program on "Condition Monitoring Control and Protection of Electrical Apparatus" (first of its kind in the country) was launched in June, 2009 as a collaborative endeavor between the TIFAC CORE and EED of NIT Hamirpur. Admissions in the program are normally made through Centralized Counseling Procedure for M.Tech Admission called CCMT with valid GATE score on all India basis. Some Ph.D. research scholars are also pursuing their research in the nodal themes related with TIFAC CORE. Research emphasis is given in industry need based identified topics for developing knowledge in thematic areas of power transformer diagnosis. Till July 2021, 149 Students of M tech (CMEA) & 05 Ph.D. students have passed out in thematic field of TIFAC CORE and serving in leading organizations. One of our passed out Ph D scholar, Dr. Mohan is presently doing Post Doctoral Research at UQAC Canada in areas relevant to TIFAC- CORE NIT Hamirpur under the guidance of eminent Prof. Fofona & another passed out student Dr. Hasmat M. is pursuing Post Doctoral in Singapore. Similarly, many other alumni of M tech Program on Condition Monitoring are serving as Faculty in NITs, IITs, and many are pursuing Phd in NITs/IITs. Some alumni are serving in Public & Pvt. Sector Utilities as well eminent organization /institutes as well.

MAJOR RESEARCH AREAS

- Ageing of solid insulations and improvements in materials used for manufacturing the power transformers.
- Detection, location of faults, Impact of fault in form of winding deformation due to short-circuit/ transportation/ageing
- Partial Discharge Measurement applications in fault diagnosis.
- Transformer Condition Monitoring practices at Site
- Performance evaluation of Transformers Feeding Steel Industry
- Insulation Oil- Paper Chemistry
- Application of Wireless Based Technologies for Prevention of Catastrophic Failures
- Failure Analysis of Power Transformers
- RLA studies on Power Transformers

FUTURE PLANS

- To go in NABL accreditation in respect of TIFAC – CORE linked laboratories.
- To liaison world class Centres/ Laboratories/ Institutes for promoting R & D in emerging areas of condition assessment of power apparatus.
- To interact with other state utilities/ Nigams for partnering with TIFAC – CORE.
- To explore opportunities for getting industry sponsored projects.
- To provide ample employment opportunities to students of PG program “Condition Monitoring Control and Protection of Electrical Apparatus” in leading industries.
- To offer more and more training programs on modules relevant to industry/ electrical utilities for safer/ upkeep of electrical substation equipment's especially transformers.
- To develop learning material in area of R & D.
- To promote collaborations with leading High Voltage test labs / Institutes on mutual benefit basis.

MAJOR FACILITIES AT TIFAC CENTRE NIT HAMIRPUR

- **M 4100 Insulation Analyzer:** Measures capacitance, Power factor and Watt losses in the solid/liquid insulation of power transformer. Also measures leakage reactance turns ratio, excitation current etc of in service power transformers and power generator analysis.
- **M 5200 Sweep Frequency Response Analyzer:** Utilized to detect the mechanical failure or movement of windings and core during short circuits, transportation, Re- location of Transformers.
- **Dirana:** To analyze the moisture of whole power transformer insulation, Dielectric properties of insulation within a wide frequency band.
- **High Performance Liquid Chromatography:** To Measure the furans in the insulating oil.
- **Thermal Imaging Camera:** To capture the thermal images of power transformer while in operation.
- **Flash Point:** To measure the flash point of insulating oil.
- **Vib Scanner:** To measure the vibration of power transformer.
- **Fourier Transform Infrared spectrophotometer:** To measure the oxidation inhibitors in the insulating oil.
- **ADTR-2K, Automatic Capacitance and Tan Delta Test Set:** Measures the capacitance dielectric constant tan delta and resistivity of insulation oil.
- **Columetric Karl Fischer Titrator:** Measure the water contents in the insulating oil.
- **Breakdown Voltage Oil Test Set:** Measures the BDV of the insulating oil.
- **Interfacial Tension Analyzer:** Measure the Interfacial Tension of insulation oil.
- **Sludge and Sediments:** To measure the sludge and sediments of insulating oil.
- **Domino Moisture in oil Analyzer:** Measures the water /moisture contents in the insulating oil.
- **Portable Dissolved Gas Analyzer:** Measures the dissolved gases in the insulating oil.
- **Acidity Titrator:** To measure the acid contents in the insulating oil.
- **Combined TGA,DTA, DSC:** Measures the thermal changes , oxidation induction and identification of paper insulation degradation.
- **UV/VIS Spectrophotometer:** Measures the dissolved decay contents of insulating liquids.
- **Solid Breakdown Voltage Test Set:** Measures the BDV of the insulating paper.

LIST OF OUR CLIENTS

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| <ul style="list-style-type: none"> ➤ M/s Himachal State Electricity Board Ltd., Shimla (H.P.). ➤ M/s Larji Power House, Kullu (H.P.). ➤ M/s JP Enterprises, Samirpur, Hamirpur (H.P.). ➤ M/s Voith Hydro Pvt Ltd., A-20 & 21, Sector 59, NOIDA (U.P.) 201 301. ➤ M/s Everest Power Pvt. Ltd., Mallana –II , Hydro Electric Project, Kullu (H.P.). ➤ M/s Faiveley Transport India Ltd. Plot No 3 Katha Bhatoli, Baddi (H.P.). ➤ M/s Crest steel & power pvt. Ltd., VPO Kurluhi Tehsil Amb, Una (H.P.). ➤ M/s Ganvi house division, Jeori (H.P.) ➤ M/s Baner power house division, Jia (H.P.) ➤ Akal college of basic science, Distt: Sirmour (H.P.) ➤ M/s Peer Panchal Pvt. Ltd. J & K. ➤ M/s Himachal Pradesh Power Corporation Ltd. Shimla | <ul style="list-style-type: none"> ➤ M/s NTPC Koldam, (H.P.) ➤ M/s Winsome Textile Industries Ltd, Industrial Area-1, Baddi (H.P.). ➤ M/s IP Enterprises Shimla ➤ M/s Luminous Industries, Una. ➤ M/s Masli Small Hydro Project (5 MW), Rohru Area, Chirgaon, Shimla (H.P.). ➤ M/s Jaypee University of Information Technology, Wagnaghat, PO Dumehar Bani, Kandaghat, Distt. Solan (H.P.). ➤ M/s Jaiprakash Sewa Sansthan WKG, Information Technology, Wagnaghat, Solan (H.P.). ➤ Himachal power transmission corporation limited, Phojal, Distt. Kullu (H.P.) ➤ M/s BVPCL Jogindernagar (H.P.) ➤ M/s Goodwill Pvt. Ltd, Rampur Shimla (HP) ➤ M/s Associated Electrical Pvt. Ltd. Haridwar |
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